

A Feature Engineering

We calculate the features for 50-point sequences, hence $N = 50$. For quantile-based features, we denote the α -quantile of a signal as Q_α . To capture local changes in acceleration, we define the absolute incremental difference as $|\Delta a_n^j| := |a_{n+1}^j - a_n^j|$. For the frequency-domain features, we denote by FFT the fast Fourier transform introduced by Nussbaumer, [6].

Table 2: Single-axis Time-Domain Features

Feature name	Formula/Description
Mean (μ_j)	$\mu_j = \frac{1}{N} \sum_{n=1}^N a_n^j$
Variance (σ_j^2)	$\sigma_j^2 = \frac{1}{N-1} \sum_{n=1}^N (a_n^j - \mu_j)^2$
Standard Deviation (σ_j)	$\sigma_j = \sqrt{\sigma_j^2}$
Root Mean Square (RMS)	$\sqrt{\frac{1}{N} \sum_{n=1}^N (a_n^j)^2}$
Shape Factor	$SF = \frac{RMS}{ \mu_j }$
Maximum	$\max_{n \in \{1, \dots, N\}} a_n^j$
Minimum	$\min_{n \in \{1, \dots, N\}} a_n^j$
Median	$\text{median}(a^j)$
Interquartile Range (IQR)	$Q_3 - Q_1$
Signal Magnitude Area (SMA)	$\frac{1}{N} \sum_{n=1}^N a_n^j $
Skewness	$\frac{1}{N} \sum_{n=1}^N \left(\frac{a_n^j - \mu_j}{\sigma_j} \right)^3$
Kurtosis	$\frac{1}{N} \sum_{n=1}^N \left(\frac{a_n^j - \mu_j}{\sigma_j} \right)^4$
Simplified Jerk Mean	$\frac{1}{N-1} \sum_{n=1}^{N-1} \Delta a_n^j $
Simplified Jerk Std	Standard deviation of $ \Delta a_n^j $
Number of Peaks	Count of local maxima in a^j
Zero Crossings	$\sum_{n=1}^{N-1} \mathbf{1}_{\{(a_n^j - a_{n+1}^j) < 0\}}$

Table 3: Single-axis Frequency-Domain Features

Feature name	Formula/Description
Energy	$\sum_{n=1}^{N/2} \text{FFT}(a^j)_n ^2$
Dominant Frequency	$\arg \max_{n \in \{1, \dots, N/2\}} \text{FFT}(a^j)_n $
Entropy	$-\sum_n p_n \log p_n, p_n = \frac{ \text{FFT}(a^j)_n }{\sum_n \text{FFT}(a^j)_n }$
Signal Entropy	Entropy of histogram bins of a^j

For the multi-axis features we derive from the SMV $\Delta smv_n := (smv_{n+1} - smv_n)$, $\Delta smv := (\Delta smv_n)_{n \in \{1, \dots, N-1\}}$, and further $\Delta^2 smv_n := (\Delta smv_{n+1} - \Delta smv_n)$ with $\Delta^2 smv := (\Delta^2 smv_n)_{n \in \{1, \dots, N-1\}}$. Let $N\Delta$ denote the number of zero-crossings of Δsmv . We also define $\text{WelchPSD}(smv)$ to be the power spectral density derived from Welch's method and the mean $\mu_{smv} = \frac{1}{N} \sum_{n=1}^N smv_n$, as well as $L = \sum_{n=1}^{N-1} |smv_{n+1} - smv_n|$ and $d = \max_{n \in \{0, \dots, N-1\}} |smv_n - smv_0|$.

For orientation-based features, such as various angles, velocity, and position, it is important to note that they still include the gravitational component and are hence not gravity-corrected. This is because the orientation of the sensor cannot be determined, as gyroscope and magnetometer data from the inertial sensor are not available.

Table 4: Time- and Frequency Domain Multi-Axis Features

Feature Name	Formula / Description
Total Mean (Sum)	$\mu_x + \mu_y + \mu_z$
Total Mean (Euclidean)	$\sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2}$
Signal Magnitude (Signed)	$\frac{1}{N} \sum_{n=1}^N (a_n^x + a_n^y + a_n^z)$
Signal Magnitude (Absolute)	$\frac{1}{N} \sum_{n=1}^N (a_n^x + a_n^y + a_n^z)$
Axis Correlation	$\rho(a^x, a^y), \rho(a^x, a^z), \rho(a^y, a^z)$
Axis Covariance	$\text{cov}(a^x, a^y), \text{cov}(a^x, a^z), \text{cov}(a^y, a^z)$
Spectral Entropy	$-\sum_i p_n \log p, p_n = \text{WelchPSD}(smv_n)$
Min / Max SMV	$\min(sm v), \max(sm v)$
Peak-to-Peak (SMV)	$\max(sm v) - \min(sm v)$
IQR (SMV)	$Q_3(sm v) - Q_1(sm v)$
Std (SMV) (σ_{smv})	$\sigma_{smv} = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (smv_n - \mu_{smv})^2}$
Skewness (SMV)	$\frac{1}{N} \sum_{n=1}^N \left(\frac{smv_n - \mu_{smv}}{\sigma_{smv}} \right)^3$
Kurtosis (SMV)	$\frac{1}{N} \sum_{n=1}^N \left(\frac{smv_n - \mu_{smv}}{\sigma_{smv}} \right)^4$
Hjorth Mobility	$\text{Mobility}(smv) = \sqrt{\frac{\text{Var}(\Delta smv)}{\text{Var}(smv)}}$
Hjorth Complexity	$\sqrt{\frac{\text{Var}(\Delta^2 smv)}{\text{Var}(\Delta smv)}} \cdot \sqrt{\frac{1}{\text{Mobility}(smv)}}$
Mean Crossing Rate	Prop. of sign changes across μ_{smv}
Differential Entropy	$0.5 \cdot \log(2\pi e \cdot \text{Var}(smv))$
Petrosian Fractal Dim.	$\frac{\log_{10}(N)}{(\log_{10}(N) + \log_{10}(\frac{N}{N+0.4N\Delta}))}$
Katz Fractal Dim.	$\frac{\log_{10}(N)}{(\log_{10}(\frac{d}{L}))}$
Tilt Angle	$\arccos\left(\frac{\mu_z}{\sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2}}\right)$
Pitch	$\arctan 2(-\mu_x, \sqrt{\mu_y^2 + \mu_z^2})$
Roll	$\arctan 2(\mu_y, \mu_z)$
Position ($pos_{j \in \{x, y, z\}}$)	Double integration of (a^x, a^y, a^z)
Velocity of Acceleration	Sum of velocity from accel.
Position of Acceleration	Sum of position from velocity

In Figure 8, the feature importances computed by the LightGBM model are visualized. The plot shows that both single- and multi-axis features, as well as features from the time and frequency domains, contribute to the model's predictive performance.

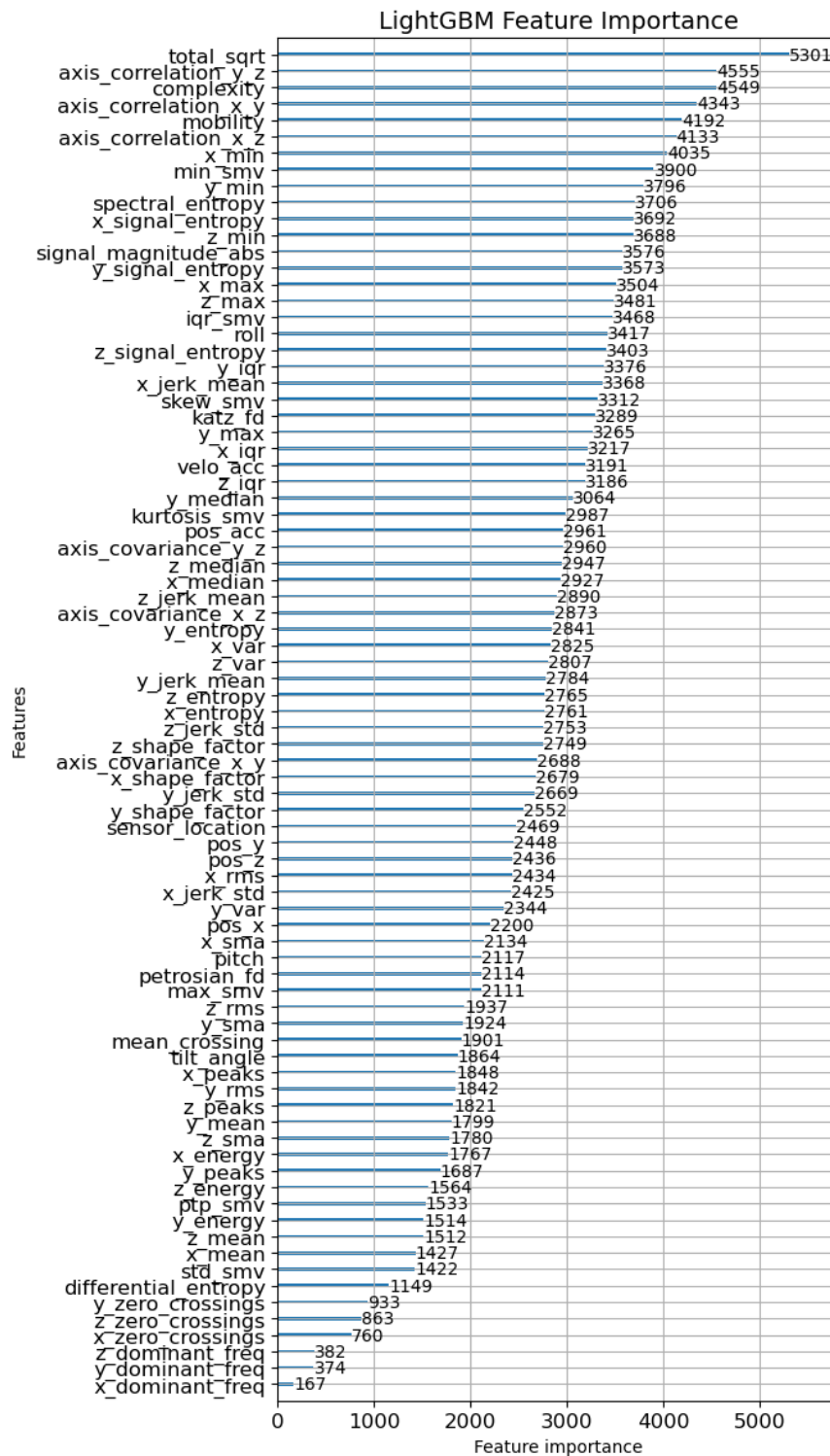


Figure 8: Feature importances of a LightGBM model.